Chapter 5, Problem 49P

Problem

Design a difference amplifier to have a gain of 4 and a common-mode input resistance of 20 kΩ at each input.

Step-by-step solution

Step 1 of 6

Draw the difference amplifier circuit diagram as shown in Figure 1.

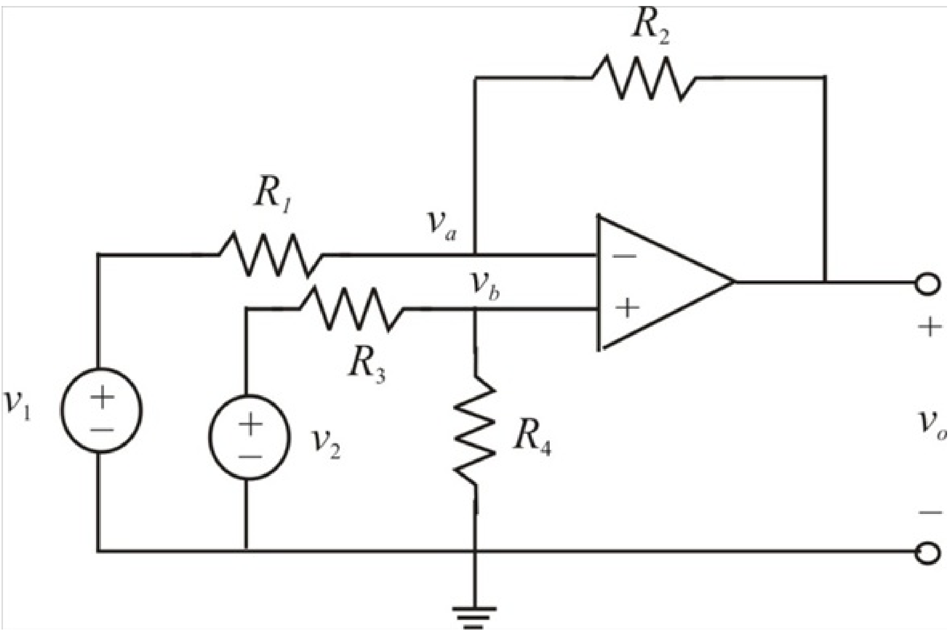

Figure 1

Step 2 of 6

The circuit is labeled as shown in Figure 2.

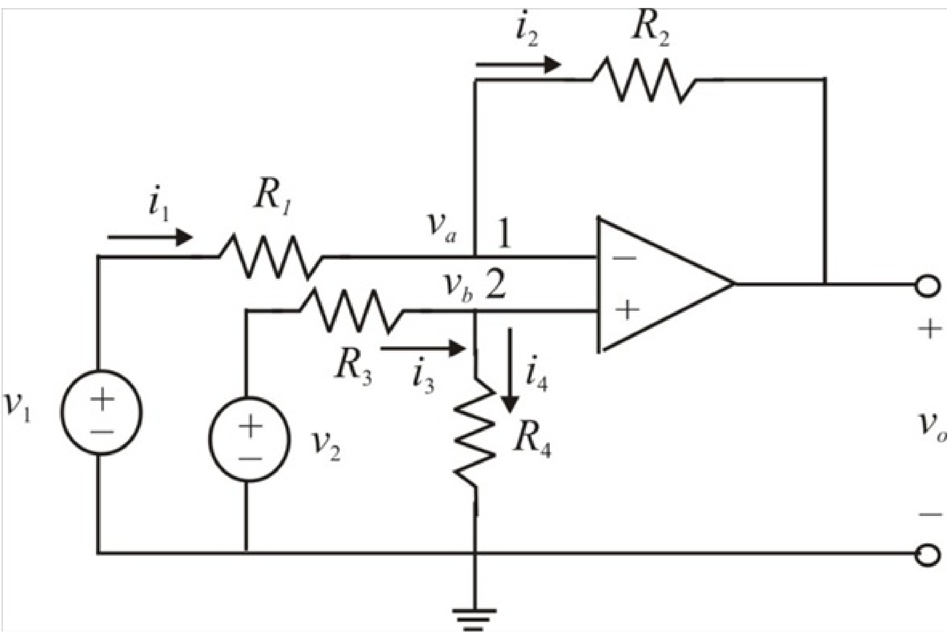

Figure 2

Step 3 of 6

The gain of the difference amplifier $= 4$

The common-mode input resistance $= 20\,\text{k}\Omega$

The currents $i_1 \text{ and } i_2$ entering in to the op amp is zero for an ideal amplifier.

Applying Kirchhoff's current law to node 1,

$$i_1 = i_2$$
$$\frac{v_1 - v_a}{R_1} = \frac{v_a - v_o}{R_2}$$
$$v_o = \left(\frac{R_2}{R_1} + 1\right)v_a - \frac{R_2}{R_1}v_1$$

Therefore $v_o = \left(\frac{R_2}{R_1} + 1\right)v_a - \frac{R_2}{R_1}v_1$ …… (1)

Applying Kirchhoff's current law to node 2,

$$i_3 = i_4$$
$$\frac{v_2 - v_b}{R_3} = \frac{v_b - 0}{R_4}$$
$$v_b = \frac{R_4}{R_3 + R_4}v_2$$

Therefore $v_b = \frac{R_4}{R_3 + R_4}v_2$ …… (2)

Step 4 of 6

But $v_a = v_b$ for an ideal op amp, substituting the equation (2) in equation (1), we get

$$\begin{aligned} v_o &= \left(\frac{R_2}{R_1} + 1\right)\frac{R_4}{R_3 + R_4}v_2 - \frac{R_2}{R_1}v_1 \\ &= \frac{R_2\left(1 + R_1 / R_2\right)}{R_1\left(1 + R_3 / R_4\right)}v_2 - \frac{R_2}{R_1}v_1 \end{aligned}$$

Since a difference amplifier must reject a signal common to the two inputs, the amplifier must have the property that $v_o = 0$ when $v_1 = v_2$.

This property exist when

$$\frac{R_1}{R_2} = \frac{R_3}{R_4}$$

Step 5 of 6

When the op amp circuit is a difference amplifier v_o becomes,

$$v_o = \frac{R_2}{R_1}(v_2 - v_1)$$

Hence the gain is

$$\frac{v_o}{v_2 - v_1} = \frac{R_2}{R_1} = 4$$
$$R_2 = 4R_1$$

But

$$\frac{R_1}{R_2} = \frac{R_3}{R_4} = \frac{1}{4}$$
$$R_4 = 4R_3$$

If $R_1 = 20\,\text{k}\Omega$, then the resistance R_2 is:

$$\begin{aligned} R_2 &= 4R_1 \\ &= 4\left(20\times10^3\right) \\ &= 80\,\text{k}\Omega \end{aligned}$$

Therefore the resistance R_2 is $\boxed{80\,\text{k}\Omega}$.

If $R_3 = 20\,\text{k}\Omega$, then the resistance R_4 is:

$$\begin{aligned} R_4 &= 4R_3 \\ &= 4\left(20\times10^3\right) \\ &= 80\,\text{k}\Omega \end{aligned}$$

Therefore the resistance R_4 is $\boxed{80\,\text{k}\Omega}$.

Step 6 of 6

The desired difference amplifier circuit diagram as shown in Figure 3.

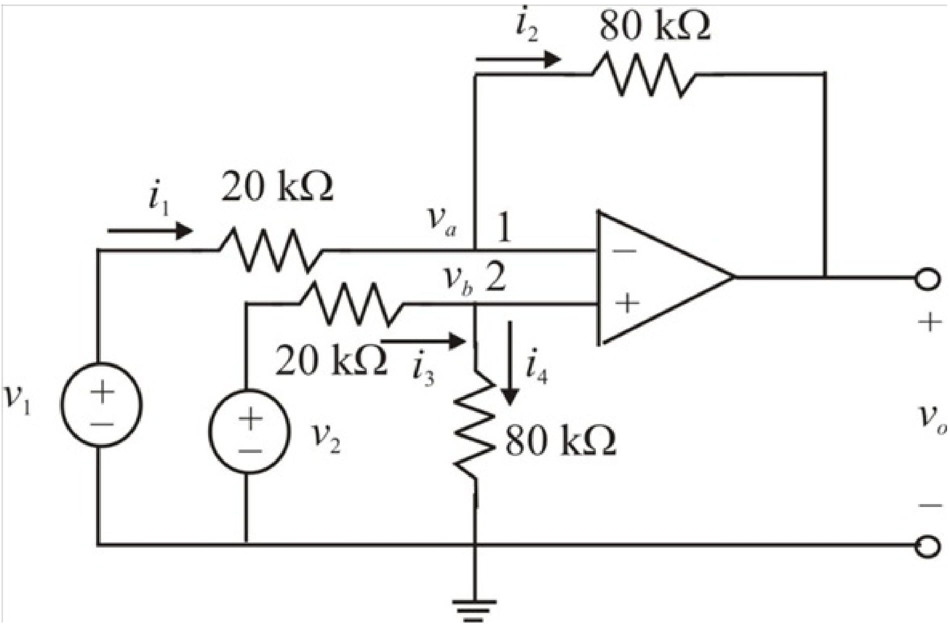

Figure 3